

LLM-informed Warm Start

Battery-domain Knowledge

battery context



MCC protocol structure

$$\theta = [I_1, \dots, I_K, \Delta s_1, \dots, \Delta s_{K-1}]^T$$

$$f(\theta) = [t_c(\theta), T_p(\theta), Q_s(\theta)]^T$$

Operating practice

$$I_1 \geq I_2 \geq I_3$$

$$\Delta s_1 + \Delta s_2 \leq 0.65$$

SOC

plating risk



LLM reasoning

Propose 15 diverse 3-stage protocols for 0–80% SOC

"high current at low SOC is safe; taper toward full charge spread candidates across trade-offs: fast/cool/long-life"

Initial Protocols

LLM output: $\theta_1, \theta_2, \dots, \theta_{15}$

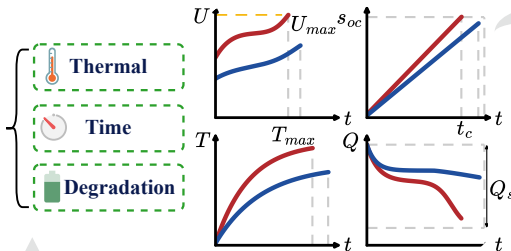
candidate screening & selection

$$D_0 = \{(\theta_j, f(\theta_j))\}_{j=1}^{n_0}, n_0 = 6$$

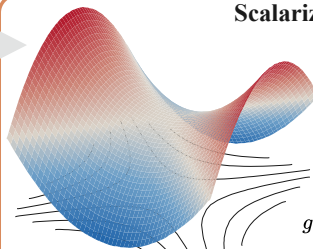
P_0 = feasible Pareto set of D_0

LLMBO-MO Loop

Battery Simulation



Scalarized GP



$$w^{(t)} \in W$$

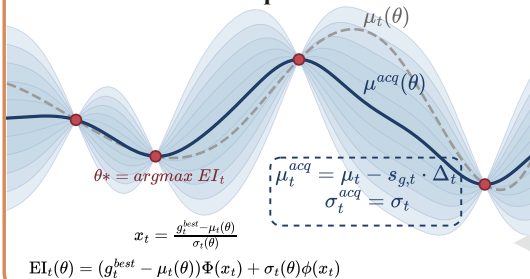
$$\bar{f} = \text{normalize}(f)$$

$$g_t^{\text{aug}}(\theta) = \rho_{\text{aug}} \sum_{i=1}^3 w_i^{(t)} \bar{f}_{i,t}(\theta)$$

$$g_t(\theta) = \max_{i=1}^3 \{w_i^{(t)} \bar{f}_{i,t}(\theta)\} + g_t^{\text{aug}}(\theta)$$

$$g(\theta) \sim \mathcal{N}(0, k(\theta; \theta')) \rightarrow (\mu_t, \sigma_t)$$

EI Acquisition



LLM Regional Guidance

"high early current saves time; keep Δs_1 moderate to limit plating risk. Propose $R_t : I_1 \in [4.1, 5.1]A$, $\Delta s_1 \in [0.24, 0.33], \hat{c}_t = 0.7$ "



$$\{R_t, \hat{c}_t\} = LLM(w^{(t)}, D_t), t \leq T_L$$

$$\lambda_t \propto \hat{c}_t \max(1 - \frac{t-1}{T_L}, 0)$$

$$\Delta_t(\theta) = \text{clip}(\lambda_t \cdot \frac{\text{cor}(\hat{g}(\theta), \hat{g}_{R_t})}{\sqrt{\text{Var}(\hat{g}_{R_t})}})$$

LLM preference density

